

AI and Machine Learning Notes

Artificial Intelligence

Artificial Intelligence, or AI, is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence. Examples include understanding language, recognizing images, making decisions, and planning.

Machine Learning

Machine Learning is a subset of AI in which a computer learns patterns from data instead of being explicitly programmed for every rule. Supervised learning uses labeled data, while unsupervised learning finds patterns in unlabeled data.

Deep Learning

Deep Learning is a branch of machine learning based on neural networks with multiple layers. Deep learning is widely used for image recognition, speech processing, natural language processing, and time-series prediction.

Neural Networks

A neural network contains layers of interconnected neurons. Each neuron receives inputs, applies weights and a bias, and passes the result through an activation function. During training, the model adjusts its weights to reduce the loss.

Natural Language Processing

Natural Language Processing, or NLP, enables computers to process and understand human language. Common tasks include text classification, sentiment analysis, translation, summarization, question answering, and information extraction.

Transformers

Transformers are neural network architectures that use attention mechanisms to process relationships between tokens. Self-attention allows a model to determine which tokens are important when representing another token. Transformers are the foundation of many modern language models.

Large Language Models

Large Language Models, or LLMs, are neural networks trained on very large collections of text. A common training objective is next-token prediction. During generation, the model predicts tokens sequentially based on the preceding context.

Embeddings

An embedding converts text or another object into a numerical vector. Similar meanings tend to have vectors that are close in the embedding space. Embeddings are useful for semantic search, recommendations, clustering, and retrieval-augmented generation.

Vector Databases

A vector database or vector index stores embeddings and supports similarity search. FAISS is a library for efficient similarity search over dense vectors. Other vector database technologies include Qdrant, Pinecone, Weaviate, Milvus, and Chroma.

Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, combines information retrieval with an LLM. Documents are split into chunks, converted into embeddings, and indexed. When a user asks a question, the question is embedded and similar document chunks are retrieved. Those chunks are provided to the LLM as context for generating the answer.

RAG Pipeline

A typical RAG pipeline is: document ingestion, text extraction, chunking, embedding generation, vector indexing, query embedding, similarity search, context construction, and LLM generation. Good chunking and retrieval are important because poor retrieval can lead to incomplete or incorrect answers.

AI Knowledge Assistant

An AI Knowledge Assistant can allow a user to upload a PDF and ask questions about its contents. The backend extracts the PDF text, creates chunks and embeddings, stores them in a FAISS index, retrieves relevant chunks for each question, and sends the retrieved context to an LLM. A FastAPI backend can expose the system through HTTP endpoints, while a React frontend can provide a chat interface.